



**CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE
PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	PHYSICAL CHEMISTRY V- ADVANCED TOPICS IN SOLID STATE AND ELECTROCHEMISTRY				
Type of Course	MAJOR /MINOR				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Preliminary ideas about structure and bonding in solids, Physical properties of solids, Dynamic electrochemistry, Solid surfaces: Adsorption and heterogenous catalysis It is desirable for the students to familiarise with the previous physical and theoretical chemistry courses, CHE2CJ102, CHE3CJ201, CHHE4CJ205, CHE5CJ301, CHE6CJ306, CHE7CJ401, and CHE7CJ401				
Course Summary	Physical properties of solids are intriguing and they are of huge technological interest. In fact, our everyday life in the modern times is intimately connected to these exciting solid materials. First two modules of this course are designed to appreciate the science of structure-property relations in solids. The third module deals with the kinetics of electrochemical processes and basic idea of some electroanalytical methods. The fourth module gives a deeper insight to the importance of surface of solids in heterogeneous catalysis.				

Course Outcomes (CO): .

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To understand the importance of structure and bonding in solids	U	C	Test /Seminar/Assignment
CO2	To analyse and correlate the structure with various physical properties in solids	Ap	P	Test /Seminar/Assignment
CO3	To comprehend the concepts of equilibrium electrochemistry	U	F	Test /Seminar/Assignment

CO4	To apply the knowledge of electrode kinetics in electrochemical processes and electroanalytical techniques	Ap	P	Test /Seminar/Assignment
CO5	To understand theory of multilayer adsorption of molecules on solid surfaces	U	C	Test /Seminar/Assignment
CO6	To apply the knowledge of adsorption for the development of heterogeneous catalysts	Ap	P	Test/Labwork/Viva
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	STRUCTURE AND BONDING IN SOLIDS		12	26
	1	Ionic bonding, radius ratio rules and structure of simple Ionic solids (NaCl, KCl, CsCl, ZnS, NiAs, CaF ₂ etc.), Partial covalent bonding in solids, Perovskite and Spinel-type structures	2	
	2	Qualitative MO diagram of hypothetical cyclic molecules of Hydrogen- H ₂ , H ₄ , H ₅ , H ₆ , H _n . Orbital interactions in solids and Band theory as applied to hypothetical one dimensional H-atom crystal, Brillouin zone, Band dispersion curves, Density of States (DOS), The Fermi level	2	
	3	Band width and nature of band dispersion and DOS in 1D H-atom crystal with varying H-H distances, Peirls distortion, Crystal orbital overlap population,	2	
	4	Band dispersion curves of 1D- chain of <i>p</i> - and <i>d</i> orbitals (1D-chain of eclipsed PtL ₄ complexes), Band theory extended to two dimensions- 1s orbitals, sigma and pi-interactions of 2 <i>p</i> orbitals	2	
	5	Band structure of 3D solids: Qualitative idea of band gap,, direct and indirect band gaps, metals, insulators and semiconductors, Ohm's law, definition of resistivity and conductivity of solids, Topological Insulators (basic idea only)	2	
	6	Bandwidth and its slope, electrical conductivity in solids and charge carriers, electrons and holes, mobility of charge carriers, charge carrier concentration, effective mass, concept of polarons, Structure-property relation in solids	2	
II	PHYSICAL PROPERTIES OF SOLIDS		12	26
	7	Semiconductors: p- and n-type doping, transistors- Photovoltaic effect and Solar energy conversion	3	

		materials, Examples: Si, CuInSe ₂ , and Methylammonium lead bromide, Solar cells Thermoelectric materials for heat to electricity direct conversion, Seebeck and Peltier effects, Examples: Bi ₂ Te ₃ , PbTe		
	8	Dielectrics, Ferroelectrics, Ferroelectricity in BaTiO ₃ , Piezoelectrics, Piezoelectricity in (Pb,Zr)TiO ₃ , Transducers	2	
	9	Optical properties of solids: Luminescence and phosphors, Lasers- Ruby Laser, Semiconducting lasers, Light emitting diodes (LED)	2	
	10	Magnetic materials: Theory and examples of Ferromagnetic, Antiferromagnetic, and Ferrimagnetic materials, Classification of Hard and Soft magnets with examples, their crystal structures and their uses, Ferrites, Nd ₂ Fe ₁₄ B, SmCo ₅ , Multiferroics and examples	2	
	11	Superconductivity, BCS theory, Critical temperature and critical field, Type-1 and Type-2 superconductors, Meissner effect, Oxide-based superconductors.	3	
III	DYNAMIC ELECTROCHEMISTRY		18	30
	12	The nature of electrolytes, Ion activity, Ion-ion and ion-solvent interaction, The electrical potential in the vicinity of an ion- Ionic thickness.	2	
	13	The Debye-Hückel equation (derivation), Limiting and extended forms of the Debye- Hückel equation, Applications of the Debye-Hückel equation to calculate the effect of ionic strength on ion reaction rates in solution - Primary and secondary salt effect	3	
	14	Electrical double layer: Helmholtz -Perrin theory, Gouy Chapman Model and Stern theory. Electrokinetic phenomena – zeta potential	2	
	15	Electrode kinetics of electrode processes, Overpotential, the Butler-Volmer equation-The relationship between current density and overvoltage, the Tafel equation.	3	
	16	Polarization: electrolytic polarization, dissolution and deposition potentials, concentration polarization	2	
	17	Determination of hydrogen overvoltage and oxygen overvoltage. Metal deposition over voltage, Principles of Polarography- the half-wave potential	2	
	18	Basic idea of Electrocatalysis, Application of electrocatalysis in Hydrogen Evolution Reactions (HER)	2	
	19	Basic principles of Galvanostatic and Potentiostatic methods in electrochemistry: Chronoamperometry,	2	

		Coulometry, Cyclic voltammetry, Chronopotentiometry, Impedance spectroscopy		
IV		SOLID SURFACES: ADSORPTION AND HETEROGENEOUS CATALYSIS	6	16
	20	Adsorption at solid surfaces: Adsorption isotherms, BET equation – derivation, Determination of surface area and pore structure of adsorbents- physical adsorption methods, X-ray methods, mercury intrusion method, chemisorption methods,	2	
	21	Features of heterogeneous catalysis: Langmuir - Hinshelwood mechanism and Eley-Rideal mechanism – illustration using the reaction $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$	2	
	22	Basic idea of experimental methods to determine surface composition of catalysts: X-ray and UV photoelectron spectroscopy (XPS, UPS), Electron energy loss spectroscopy (EELS), Surface extended X-ray absorption fine structure spectroscopy (SEXAFS)	2	
V	Open ended		12	
	1. Computer simulations of crystal structures of various structure types from available cif files, Simulation of reciprocal lattice using suitable computer programs 2. Demonstration of band dispersion curves of simple systems such as graphene, band structure calculation of Si by using Quantum Espresso or other software packages. 3. Explanation of electrical conductivity measurements of semiconductors via four-point probe method, Magnetic hysteresis in soft and hard magnets, Optical band gap by using diffuse reflectance spectroscopy 4. (Virtual lab) demonstration of Cyclic voltammetry and impedance spectroscopy 5. Adsorption experiments on activated charcoal and other solid surfaces			
	References Modules I and II 1. Solid State Chemistry and Applications by A. R. West, 2nd edition, 2014, Wiley 2. How Chemistry and Physics Meet in the Solid State, Roald Hoffmann, Angew. Chem. Int. Ed. Engl. 26 (1987) 846-878 Module III 3. Electrochemical methods: Fundamentals and Applications, by Allen J. Bard and Larry R. Faulkner, 2nd Edition			

	4. Volume 2a, Modern Electrochemistry, 2nd edition, Fundamentals of Electrodeics by John O'M Bockris, Amulya K. N. Reddy, and Maria Gamboa-Aldeco, 5. Volume 1, Modern Electrochemistry, 2nd edition, Ionics by John O'M Bockris, and Amulya K. N. Reddy Module IV 6. Physical Chemistry: Thermodynamics, Structure and Change, 10th Edition, P. Atkins and J. de Paula, (W. H Freeman and Company, New York) 7. K. Laidler, Chemical Kinetics, 3rd Ed., Pearson Education, New Delhi, 2004. Further reading 8. C. N. R. Rao and J. Gopalakrishnan, New Directions in Solid State Chemistry, 2nd Ed., Cambridge University Press, 2004. 9. Introduction to Surface Physical Chemistry, K. Christmann, Springer-Verlag, Berlin, 1991 10. Direct Energy Conversion, Andrea M. Mitofsky, 2018		
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Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	-	-	3	2	3	2	2	-	2	-	1
CO2	3	2	-	-	3	2	3	2	1	-	2	-	1
CO3	3	2	-	-	3	2	3	2	1	-	2	-	1
CO4	3	2	-	-	3	3	3	2	1	-	1	-	1
CO5	3	2	-	1	3	3	3	2	1	-	3	-	1
CO6	3	-	2	3	3	3	3	2	1	2	3	2	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Practical exam (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment /viva	Practical skill evaluation	End Semester Examinations
CO 1		✓		✓
CO 2		✓		✓
CO 3	✓			✓
CO 4	✓			✓
CO 5	✓	✓		✓
CO 6	✓	✓	✓	✓